

Analytical Distance Calculations between von Mises and Wrapped Cauchy Distributions

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1 Introduction

This document summarizes the mathematical details of the analytical distance calculations (KL divergence and 1-Wasserstein distance) between the von Mises distribution and the wrapped Cauchy distribution on the circle S^1 .

2 Distributions

Let P be the von Mises distribution with mean direction $\mu_P \in [-\pi, \pi)$ and concentration parameter $\kappa > 0$:

$$p(\theta) = \frac{1}{2\pi I_0(\kappa)} e^{\kappa \cos(\theta - \mu_P)} \quad (1)$$

where $I_n(\kappa)$ is the modified Bessel function of the first kind of order n .

Let Q be the wrapped Cauchy distribution with mean direction $\mu_Q \in [-\pi, \pi)$ and concentration parameter $\rho \in [0, 1)$:

$$q(\theta) = \frac{1 - \rho^2}{2\pi(1 + \rho^2 - 2\rho \cos(\theta - \mu_Q))} \quad (2)$$

3 Kullback-Leibler Divergence

3.1 Divergence from von Mises to Wrapped Cauchy

The KL divergence $D_{\text{KL}}(P \parallel Q)$ is defined as:

$$D_{\text{KL}}(P \parallel Q) = \int_0^{2\pi} p(\theta) \log \left(\frac{p(\theta)}{q(\theta)} \right) d\theta \quad (3)$$

Using the Fourier series expansion:

$$\log(1 + \rho^2 - 2\rho \cos x) = -2 \sum_{n=1}^{\infty} \frac{\rho^n}{n} \cos(nx) \quad (4)$$

and the trigonometric moments of the von Mises distribution:

$$E_P[\cos(n(\theta - \mu_Q))] = \frac{I_n(\kappa)}{I_0(\kappa)} \cos(n(\mu_P - \mu_Q)) \quad (5)$$

we obtain the fast-converging series representation:

$$D_{\text{KL}}(P \parallel Q) = \kappa \frac{I_1(\kappa)}{I_0(\kappa)} - \log I_0(\kappa) - \log(1 - \rho^2) - 2 \sum_{n=1}^{\infty} \frac{\rho^n I_n(\kappa)}{n I_0(\kappa)} \cos(n(\mu_P - \mu_Q)) \quad (6)$$

3.2 Divergence from Wrapped Cauchy to von Mises

The KL divergence $D_{\text{KL}}(Q \parallel P)$ can be calculated in closed-form. Using $E_Q[\cos(\theta - \mu_P)] = \rho \cos(\mu_Q - \mu_P)$ and $E_Q[\log(1 + \rho^2 - 2\rho \cos(\theta - \mu_Q))] = 2 \log(1 - \rho^2)$, we get:

$$D_{\text{KL}}(Q \parallel P) = \log I_0(\kappa) - \log(1 - \rho^2) - \kappa \rho \cos(\mu_Q - \mu_P) \quad (7)$$

4 1-Wasserstein Distance (Aligned Means)

When the mean directions are aligned ($\mu_P = \mu_Q$) and one distribution is more concentrated than the other, the 1-Wasserstein distance $W_1(P, Q)$ on the circle reduces to the difference of their mean absolute deviations on $[-\pi, \pi]$:

$$W_1(P, Q) = |E_P[|\theta|] - E_Q[|\theta|]| \quad (8)$$

4.1 Derivation and Proof

Without loss of generality, assume the aligned mean direction is $\mu_P = \mu_Q = 0$ and the distributions are defined on $[-\pi, \pi]$. Since both the von Mises and wrapped Cauchy distributions are symmetric around $\theta = 0$, their cumulative distribution functions (CDFs) satisfy $F_P(-\theta) = 1 - F_P(\theta)$ and $F_Q(-\theta) = 1 - F_Q(\theta)$ for $\theta \in [0, \pi]$.

The 1-Wasserstein distance on the circle S^1 is given by:

$$W_1(P, Q) = \min_{\alpha \in \mathbb{R}} \int_{-\pi}^{\pi} |F_P(\theta) - F_Q(\theta) - \alpha| d\theta \quad (9)$$

Let $D(\theta) = F_P(\theta) - F_Q(\theta)$. By the symmetry of the CDFs, we have:

$$D(-\theta) = (1 - F_P(\theta)) - (1 - F_Q(\theta)) = -D(\theta) \quad (10)$$

which implies that the difference function $D(\theta)$ is odd. The optimal shift α^* that minimizes the integral is the median of $D(\theta)$ over $[-\pi, \pi]$. Since $D(\theta)$

is a continuous odd function on the symmetric interval $[-\pi, \pi]$, its median is exactly 0. Therefore, $\alpha^* = 0$, and the distance simplifies to:

$$W_1(P, Q) = \int_{-\pi}^{\pi} |F_P(\theta) - F_Q(\theta)| d\theta \quad (11)$$

Since $|D(\theta)|$ is an even function, we can restrict the integration to $[0, \pi]$:

$$W_1(P, Q) = 2 \int_0^{\pi} |F_P(\theta) - F_Q(\theta)| d\theta \quad (12)$$

Now, we examine the condition under which the CDF difference $D(\theta) = F_P(\theta) - F_Q(\theta)$ does not change sign (i.e., has 0 crossings) on the open interval $(0, \pi)$. For symmetric, unimodal circular distributions centered at 0, the derivative $D'(\theta) = p(\theta) - q(\theta)$ can change sign at most twice in $(0, \pi)$. Since $D(0) = D(\pi) = 0$, the derivative must change sign at least once.

A necessary and sufficient condition for $D(\theta)$ to have no crossings in $(0, \pi)$ is:

$$(p(0) - q(0))(p(\pi) - q(\pi)) \leq 0 \quad (13)$$

Indeed:

- If $(p(0) - q(0))(p(\pi) - q(\pi)) \leq 0$, the derivative $D'(\theta)$ changes sign exactly once in $(0, \pi)$. Thus, $D(\theta)$ is unimodal and remains strictly positive (if $p(0) > q(0)$) or strictly negative (if $p(0) < q(0)$) on $(0, \pi)$, meaning it has 0 crossings.
- If $(p(0) - q(0))(p(\pi) - q(\pi)) > 0$, the derivative $D'(\theta)$ changes sign exactly twice in $(0, \pi)$. By Rolle's theorem and the Intermediate Value Theorem, this implies that $D(\theta)$ must cross 0 exactly once in $(0, \pi)$.

Therefore, if the condition $(p(0) - q(0))(p(\pi) - q(\pi)) \leq 0$ is satisfied, $F_P(\theta) - F_Q(\theta)$ does not change sign on $(0, \pi)$. Assuming without loss of generality that $F_P(\theta) \geq F_Q(\theta)$ for all $\theta \in [0, \pi]$, we can remove the absolute value:

$$W_1(P, Q) = 2 \int_0^{\pi} (F_P(\theta) - F_Q(\theta)) d\theta \quad (14)$$

Next, we establish the relationship between this integral and the mean absolute deviation (MAD). The MAD of a symmetric distribution P on $[-\pi, \pi]$ is defined as:

$$E_P[|\theta|] = \int_{-\pi}^{\pi} |\theta| p(\theta) d\theta = 2 \int_0^{\pi} \theta p(\theta) d\theta \quad (15)$$

Using integration by parts with $u = \theta$ and $dv = p(\theta)d\theta$ (where $v = F_P(\theta) - 1$), we obtain:

$$\begin{aligned} 2 \int_0^\pi \theta p(\theta) d\theta &= 2 [\theta(F_P(\theta) - 1)]_0^\pi - 2 \int_0^\pi (F_P(\theta) - 1) d\theta \\ &= 2\pi(1 - 1) - 0 + 2 \int_0^\pi (1 - F_P(\theta)) d\theta \\ &= 2 \int_0^\pi (1 - F_P(\theta)) d\theta \end{aligned} \quad (16)$$

Thus, we have:

$$E_P[|\theta|] = 2 \int_0^\pi (1 - F_P(\theta)) d\theta, \quad E_Q[|\theta|] = 2 \int_0^\pi (1 - F_Q(\theta)) d\theta \quad (17)$$

Substituting these expressions back into the simplified 1-Wasserstein integral yields:

$$\begin{aligned} W_1(P, Q) &= 2 \int_0^\pi (F_P(\theta) - F_Q(\theta)) d\theta \\ &= 2 \int_0^\pi (1 - F_Q(\theta)) d\theta - 2 \int_0^\pi (1 - F_P(\theta)) d\theta \\ &= E_Q[|\theta|] - E_P[|\theta|] \end{aligned} \quad (18)$$

Taking the absolute value to handle both cases where either P or Q is more concentrated, we arrive at the desired result:

$$W_1(P, Q) = |E_P[|\theta|] - E_Q[|\theta|]| \quad (19)$$

4.2 Analytical Expressions for Symmetric Distributions

The mean absolute deviation of each symmetric distribution is given by:

$$E_P[|\theta|] = \frac{\pi}{2} - \frac{4}{\pi} \sum_{k=0}^{\infty} \frac{I_{2k+1}(\kappa)}{(2k+1)^2 I_0(\kappa)} \quad (20)$$

$$E_Q[|\theta|] = \frac{\pi}{2} - \frac{4}{\pi} \sum_{k=0}^{\infty} \frac{\rho^{2k+1}}{(2k+1)^2} \quad (21)$$

Thus, the 1-Wasserstein distance is:

$$W_1(P, Q) = \frac{4}{\pi} \left| \sum_{k=0}^{\infty} \frac{\rho^{2k+1} - \frac{I_{2k+1}(\kappa)}{I_0(\kappa)}}{(2k+1)^2} \right| \quad (22)$$